

Nowcasting Macroeconomic Variables with a Sparse Mixed Frequency Dynamic Factor Model – Online Appendix

Domenic Franjic*

Karsten Schweikert†

University of Hohenheim

University of Hohenheim

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A Proofs

Throughout this section, we consider the following notation. As in the main text, a matrix $\mathbf{A}$ of dimensions $N \times M$ and a column vector $\mathbf{a}$ of length N are denoted as $\mathbf{A} = (a_{n,m})_{n=1,m=1}^{N,M}$ and $\mathbf{a} = (a_n)_{n=1}^N$, respectively. The l_p -norm of a vector $\mathbf{a} = (a_n)_{n=1}^N$ is defined as $\|\mathbf{a}\|_p := \left(\sum_{n=1}^N |a_n|^p\right)^{\frac{1}{p}}$ for $p \in \mathbb{N}$. For a symmetric Matrix $\mathbf{A} = (a_{n,m})_{n=1,m=1}^{N,N}$, $\bar{\sigma}(\mathbf{A})$ and $\underline{\sigma}(\mathbf{A})$ denote the maximum and minimum eigenvalues, respectively. The Frobenius-Norm of a matrix $\mathbf{A}$ is denoted as $\|\mathbf{A}\|_F := \sqrt{\text{tr}(\mathbf{A}'\mathbf{A})}$ and the induced Euclidean norm as $\|\mathbf{A}\| := \sqrt{\bar{\sigma}(\mathbf{A}'\mathbf{A})}$. The $N \times N$ identity matrix is given by $\mathbf{I}_N$. For a given matrix $\mathbf{A}$, the convention is used that $\mathbf{A} = \mathcal{O}_P(s_n)$ if $\|\mathbf{A}\| = \mathcal{O}_P(s_n)$ for some sequence $(s_n)_{n \in \mathbb{N}}$. Lastly, for convenience, we define $\mathbf{G} := T^{-1}\mathbf{X}\mathbf{X}'$, where $\mathbf{X} = (x_{n,t})_{n=1,t=1}^{N,T}$ is the data matrix. For simplicity, we employ the convention also used in the main text and use integer time indices $t = 1, 2, \dots, T$ instead of the multiples-of-a-third adaption.

The model under consideration is an approximate dynamic factor model as used in, for example, [Doz et al. \(2011\)](#). Written in state-space form, the model is formally

*Corresponding Author; Address: University of Hohenheim, Core Facility Hohenheim & Institute of Economics, Schloss Hohenheim 1 C, 70593 Stuttgart, Germany, e-mail: franjic@uni-hohenheim.de

†Address: University of Hohenheim, Core Facility Hohenheim & Institute of Economics, Schloss Hohenheim 1 C, 70593 Stuttgart, Germany, e-mail: karsten.schweikert@uni-hohenheim.de

expressed as

$$\begin{aligned} \mathbf{x}_t &= \mathbf{\Lambda} \mathbf{f}_t + \boldsymbol{\xi}_t, \text{ for } t = 1, 2, \dots \\ \Phi^P(\mathbb{L}) \mathbf{f}_t &= \boldsymbol{\epsilon}_t \\ \Psi^Q(\mathbb{L}) \boldsymbol{\xi}_t &= \boldsymbol{\nu}_t. \end{aligned} \quad (\text{A.1})$$

Here, $\mathbf{x}_t := (x_{n,t})_{n=1}^N$ is the vector of observations at time t while $\mathbf{f}_t := (f_{r,t})_{r=1}^R$ is a vector of latent factors driving the system. In general, $0 < R \ll N$, i.e., the number of factors is considerably smaller than the number of variables. $\mathbf{\Lambda} := (\lambda_{n,r})_{n=1,r=1}^{N,R}$ is the factor loading matrix. $\Phi^P(\mathbb{L})$ and $\Psi^Q(\mathbb{L})$ are operators that map a time series vector $\mathbf{a}_t$ onto P and Q lags of itself, respectively. Formally, the matrix-polynomial-operators read

$$\begin{aligned} \Phi^P(\mathbb{L}) \mathbf{a}_t &:= \mathbf{a}_t - \Phi_1 \mathbf{a}_{t-1} - \dots - \Phi_P \mathbf{a}_{t-P} \\ \Psi^Q(\mathbb{L}) \mathbf{a}_t &:= \mathbf{a}_t - \Psi_1 \mathbf{a}_{t-1} - \dots - \Psi_Q \mathbf{a}_{t-Q}, \end{aligned}$$

where $\Phi_1, \dots, \Phi_P, \Psi_1, \dots, \Psi_Q$ are weighting matrices. Lastly, $\boldsymbol{\epsilon}$ and $\boldsymbol{\nu}$ are assumed to be i.i.d. Gaussian processes.

For the convenience of the reader, we also restate the definition of the SPCA problem by Zou et al. (2006) as is stated in the main text (see also Despois and Doz, 2023). The SPCA problem is given by

$$\min_{\mathbf{A}, \mathbf{\Lambda}} \left\{ \frac{1}{T} \|\mathbf{X}' - \mathbf{X}' \mathbf{\Lambda} \mathbf{A}'\|_F^2 + \sum_{r=1}^R \kappa_{r,1} \|\boldsymbol{\lambda}_r\|_1 + \kappa_2 \sum_{r=1}^R \|\boldsymbol{\lambda}_r\|_2^2 \right\} \text{ s.t. } \mathbf{A}' \mathbf{A} = \mathbf{I}_{R \times R}, \quad (\text{A.2})$$

where $\mathbf{\Lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_R)$ and $\mathbf{A} = (\boldsymbol{\alpha}_1, \dots, \boldsymbol{\alpha}_R)$.

Following Zou et al. (2006), this problem is solved iteratively. Consider the singular value decomposition of $\mathbf{X}'$

$$\mathbf{X}' = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0', \quad (\text{A.3})$$

where $\mathbf{U}_0 := (u_{n,r,0})_{n=1,r=1}^{N,R}$, $\mathbf{D}_0 := \text{diag}(d_{1,0}, \dots, d_{R,0})$, and $\mathbf{V}_0 := (v_{r,s,0})_{r=1,s=1}^{R,R}$, with $\mathbf{U}_0' \mathbf{U}_0 = \mathbf{V}_0' \mathbf{V}_0 = \mathbf{I}_{R \times R}$. Furthermore, define $\widehat{\mathbf{A}}_0 := (\mathbf{v}_{1,0}, \dots, \mathbf{v}_{R,0})$, where $\mathbf{v}_{r,0}$ is the vector corresponding to the r th column of $\mathbf{V}_0$. For $i = 0, 1, \dots$ repeat the following steps until convergence:

1. For $\widehat{\mathbf{A}}_i = (\widehat{\boldsymbol{\alpha}}_{1,i}, \dots, \widehat{\boldsymbol{\alpha}}_{R,i})$ and $r = 1, \dots, R$ solve the EN type problem

$$\widehat{\boldsymbol{\lambda}}_{r,i} = \arg \min_{\boldsymbol{\lambda}_{r,i}} \left\{ (\boldsymbol{\alpha}_{r,i} - \boldsymbol{\lambda}_{r,i})' \mathbf{X} \mathbf{X}' (\boldsymbol{\alpha}_{r,i} - \boldsymbol{\lambda}_{r,i}) + \kappa_2 \|\boldsymbol{\lambda}_{r,i}\|_2 + \kappa_{r,1} \|\boldsymbol{\lambda}_{r,i}\|_1 \right\}. \quad (\text{A.4})$$

2. Be $\widehat{\mathbf{\Lambda}}_i = (\widehat{\boldsymbol{\lambda}}_{1,i}, \dots, \widehat{\boldsymbol{\lambda}}_{R,i})$ and $\mathbf{U}_{i+1}$, $\mathbf{D}_{i+1}$, and $\mathbf{V}_{i+1}$, such that

$$\mathbf{X} \mathbf{X}' \widehat{\mathbf{\Lambda}}_i = \mathbf{U}_{i+1} \mathbf{D}_{i+1} \mathbf{V}_{i+1}'. \quad (\text{A.5})$$

Then set $\widehat{\mathbf{A}}_{i+1} = \mathbf{U}_{i+1} \mathbf{V}_{i+1}'$.

Assuming convergence at the I th step, retrieve the SPCA estimator of the loading matrix as

$$\hat{\mathbf{\Lambda}} = (\hat{\boldsymbol{\lambda}}_{1,I}, \dots, \hat{\boldsymbol{\lambda}}_{R,I}). \quad (\text{A.6})$$

A.1 Proof of Theorem 1

To prove Theorem 1, we build on the approach of Despois and Doz (2023, Online Appendix), drawing on their supporting propositions and lemmas. We adopt the initial structure of their proof for Theorem 1, and modify it to an adjusted convergence rate and individual penalties per factor. We do so to confirm that their supporting propositions and lemmas hold under our modification. Thereafter, we proceed by adapting their arguments to establish element-wise convergence rather than an average result. To this end, denote $e_1 \geq \dots \geq e_N$ as the ordered eigenvalues of $\mathbf{G}$ and $\mathbf{p}_1, \dots, \mathbf{p}_N$ as the corresponding eigenvectors. Further, define $\mathbf{P} := (\mathbf{p}_1, \dots, \mathbf{p}_R)$ and $\mathbf{E} := \text{diag}(e_1, \dots, e_R)$.

Consider the following Lemma about the properties of the common PCA estimator derived by Doz et al. (2011).

Lemma 1 (PCA properties) *The PCA estimator has the following properties.*

- i) $\mathbf{E} = \mathcal{O}_P(N)$
- ii) $\mathbf{E}^{-1} = \mathcal{O}_P(N^{-1})$
- iii) $\mathbf{G}^{-1} = \mathcal{O}_P(1)$

Proof. See Doz et al. (2011, Appendix) and Despois and Doz (2023, Online Appendix). ■

Next, we require Proposition 1, Proposition 2, and Lemma 2 stated below to establish Theorem 1. These are modified versions of Proposition 1, Proposition 2, and Lemma 1 in Despois and Doz (2023, Online Appendix). We follow their arguments directly, to show that these still hold under our adjusted convergence rate and unique factor penalization.

Proposition 1 *Let $\mathbf{K} := \text{diag}(\kappa_{1,1}, \dots, \kappa_{R,1})$. At iteration step i , $\hat{\mathbf{\Lambda}}_i$ can be written as:*

$$\hat{\mathbf{\Lambda}}_i = (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \left(\mathbf{G} \hat{\mathbf{A}}_i - \frac{1}{2} \mathbf{Z}_i \mathbf{K} \right),$$

where $\mathbf{Z}_i$ is a $N \times R$ matrix whose generic element satisfies $|z_{n,r,i}| \leq 1$.

Proof. From the above it is known that, at the i th iteration step, an estimator of $\hat{\mathbf{\Lambda}}_i$ is computed by solving (A.4) for each $r = 1, \dots, R$. For a given iteration step i and factor r , define the function

$$f_r(\boldsymbol{\lambda}) := (\boldsymbol{\alpha}_{r,i} - \boldsymbol{\lambda}_i)' \mathbf{X} \mathbf{X}' (\boldsymbol{\alpha}_{r,i} - \boldsymbol{\lambda}_i) + \kappa_2 \|\boldsymbol{\lambda}_i\|_2 + \kappa_{r,1} \|\boldsymbol{\lambda}_i\|_1.$$

Basic convex analysis results (see, for example, [Rockafellar, 1997](#)) then give the subdifferential of f_r in $\boldsymbol{\lambda}_r$ as

$$\partial f_r(\boldsymbol{\lambda}) = -2\mathbf{G}\boldsymbol{\alpha}_{r,i} + 2\mathbf{G}\boldsymbol{\lambda}_{r,i} + 2\kappa_2\boldsymbol{\lambda}_{r,1} + \kappa_{r,1}\mathbf{z}_{r,i},$$

where $\mathbf{z}_{r,i} = (z_{1,r,i}, \dots, z_{n,r,i})'$ with

$$\begin{cases} z_{n,r,i} = 1, & \text{if } \lambda_{n,r,i} > 0 \\ z_{n,r,i} = -1, & \text{if } \lambda_{n,r,i} < 0 \\ z_{n,r,i} \in [0, 1], & \text{if } \lambda_{n,r,i} = 0 \end{cases} \quad (\text{A.7})$$

for $n = 1, \dots, N$.

An estimator of $\boldsymbol{\lambda}_{r,i}$ is now obtained by solving

$$-2\mathbf{G}\boldsymbol{\alpha}_{r,i} + 2\mathbf{G}\boldsymbol{\lambda}_{r,i} + 2\kappa_2\boldsymbol{\lambda}_{r,i} + \kappa_{r,1}\mathbf{z}_{r,i} \stackrel{!}{=} 0$$

such that (A.7) holds. This then gives

$$\widehat{\boldsymbol{\lambda}}_{r,i} = (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \left(\mathbf{G}\boldsymbol{\alpha}_{r,i} - \frac{1}{2}\mathbf{z}_{1,r}\kappa_{r,1} \right).$$

Lastly, defining $\mathbf{Z}_i := (z_{1,i}, \dots, z_{R,i})$ and $\mathbf{K} := \text{diag}(\kappa_{1,1}, \dots, \kappa_{R,1})$, and given that $\widehat{\mathbf{A}}_i = (\mathbf{a}_{1,i}, \dots, \mathbf{a}_{R,i})$, the statement follows. ■

Proposition 2 *Assume CR1–CR4 and A1–A3 hold and $\max_{r=1,\dots,R} \kappa_{r,1} = \mathcal{O}(T^{-1/2}N^{-3/2})$. For any i , $\widehat{\mathbf{A}}_i$ can be written as*

$$\widehat{\mathbf{A}}_i = \mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1}),$$

where $\mathbf{M}_i$ is a $R \times R$ matrix with

$$\mathbf{M}_i' \mathbf{M}_i = \mathbf{I}_{R \times R} + \mathcal{O}_P(N^{-1}).$$

The proof of Proposition (2) requires the following Lemma.

Lemma 2 *Assume that $\widehat{\mathbf{A}}_i = \mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1})$, where $\mathbf{M}_i \in \mathbb{R}^{R \times R}$ is a matrix such that $\mathbf{M}_i' \mathbf{M}_i = \mathbf{I}_{R \times R} + \mathcal{O}_P(N^{-1})$. Then,*

$$i) \quad \widehat{\boldsymbol{\Lambda}}_i = \mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1}) - \frac{1}{2}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_i \mathbf{K},$$

$$ii) \quad \mathbf{G}\widehat{\boldsymbol{\Lambda}}_i = \mathbf{P}\mathbf{E}\mathbf{M}_i + \mathcal{O}_P(1).$$

Proof.

i) Assuming that $\widehat{\mathbf{A}}_i = \mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1})$ holds, it then follows from [Proposition 1](#) that

$$\begin{aligned}\widehat{\mathbf{\Lambda}}_i &= (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \left(\mathbf{G} (\mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1})) - \frac{1}{2} \mathbf{Z}_i \mathbf{K} \right) \\ &= (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G} \mathbf{P} \mathbf{M}_i + (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G} \mathcal{O}_P(N^{-1}) \\ &\quad - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_i \mathbf{K}.\end{aligned}\tag{A.8}$$

Since $\mathbf{P}$ represents the first R eigenvectors, corresponding to the eigenvalues in $\mathbf{E}$, of the matrix $\mathbf{G}$, it follows that $\mathbf{P}$ also represents the first R eigenvectors, corresponding to the eigenvalues in $(\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})$, of $(\mathbf{G} + \kappa_2 \mathbf{I}_N)$. $\mathbf{P}$ additionally represents the first R eigenvectors, corresponding to the eigenvalues in $(\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1}$, of $(\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1}$. From this follows that the first term in (A.8) can be rewritten as

$$(\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G} \mathbf{P} \mathbf{M}_i = \mathbf{P} (\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{E} \mathbf{M}_i.\tag{A.9}$$

Since $(\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{E} = \text{diag} \left(\frac{e_1}{e_1 + \kappa_2}, \dots, \frac{e_R}{e_R + \kappa_2} \right)$, and

$$\frac{e_r}{e_r + \kappa_2} \sim 1 - \frac{\kappa_2}{e_r} = 1 + \mathcal{O}_P(N^{-1}),$$

for $r = 1, \dots, R$, it follows that

$$(\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{E} = \mathbf{I}_{R \times R} + \mathcal{O}_P(N^{-1}).$$

Moreover, $\mathbf{P}'\mathbf{P} = \mathbf{I}_{R \times R}$ implies $\mathbf{P} = \mathcal{O}_P(1)$ and $\mathbf{M}_i' \mathbf{M}_i = \mathbf{I}_{R \times R} + \mathcal{O}(N^{-1})$ implies $\mathbf{M} = \mathcal{O}_P(1)$. Eq. (A.9) then reads

$$(\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G} \mathbf{P} \mathbf{M}_i = \mathbf{P} \mathbf{M}_i + \mathcal{O}_P(N^{-1}).$$

With respect to the second term in (A.8), it is obvious that $\frac{e_r}{e_r + \kappa_2} < 1$ for $r = 1, \dots, R$, where $\frac{e_r}{e_r + \kappa_2}$ is the r th eigenvalue of $(\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G}$. Consequently, it follows that

$$\begin{aligned}\|(\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G}\| &\leq 1 \\ \Rightarrow \mathcal{O}_P(N^{-1}) (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{G} &= \mathcal{O}_P(N^{-1}).\end{aligned}$$

Summarising, the above shows that

$$\widehat{\mathbf{\Lambda}}_i = \mathbf{P} \mathbf{M}_i + \mathcal{O}_P(N^{-1}) - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_i \mathbf{K},$$

which proves part i) of [Lemma 2](#).

ii) From i) follows that

$$\begin{aligned}
\mathbf{G}\widehat{\boldsymbol{\Lambda}}_i &= \mathbf{G} \left(\mathbf{P}\mathbf{M}_i + \mathcal{O}_P(N^{-1}) - \frac{1}{2}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_i\mathbf{K} \right) \\
&= \mathbf{P}\mathbf{E}\mathbf{M}_i + \mathbf{G}\mathcal{O}_P(N^{-1}) - \frac{1}{2}\mathbf{G}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_i\mathbf{K} \\
&= \mathbf{P}\mathbf{E}\mathbf{M}_i + \mathcal{O}_P(1) - \frac{1}{2}\mathbf{G}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_i\mathbf{K},
\end{aligned} \tag{A.10}$$

where the last line follows from the fact that $\mathbf{G} = \mathcal{O}_P(N)$.

Since

$$\left\| \mathbf{G}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \right\| = \left\| (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{G} \right\| \leq 1,$$

$$\|\mathbf{Z}_i\|^2 \leq \text{tr}(\mathbf{Z}_i'\mathbf{Z}_i) \leq RN \Rightarrow \|\mathbf{Z}_i\| = \mathcal{O}_P(N^{1/2}),$$

and

$$\|\mathbf{K}\| = \max_{r=1,\dots,R} \kappa_{r,1} = \mathcal{O}(T^{-1/2}N^{-3/2}),$$

it follows that

$$\left\| \mathbf{G}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_i\mathbf{K} \right\| \leq \left\| \mathbf{G}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \right\| \|\mathbf{Z}_i\| \|\mathbf{K}\| = \mathcal{O}_P(T^{-1/2}N^{-1}).$$

Inserting the above results into (A.10) proves statement ii). ■

Proof of Proposition 2.

The proof of Proposition 2 is the same as that of Proposition 2 of Despois and Doz (2023, Online Appendix), as the results of Lemma 2 and Lemma 2 of Despois and Doz (2023, Online Appendix) coincide, despite our modifications. ■

We are now able to prove Theorem 1. To do so, we draw on parts of the argument from the proof of Theorem 1 in Despois and Doz (2023, Online Appendix), and extend their reasoning by our adjusted convergence rate to establish element-wise results.

Proof of Theorem 1.

Following Despois and Doz (2023, Online Appendix), assume the SPCA algorithm to converge at the I th step and define $\widehat{\mathbf{D}} := \text{diag}(\widehat{\boldsymbol{\Lambda}}_I'\mathbf{G}\widehat{\boldsymbol{\Lambda}}_I)$. Next, from Lemma 2 it

follows that

$$\begin{aligned}
\hat{\Lambda}'_I \mathbf{G} \hat{\Lambda}_I &= \left(\mathbf{P} \mathbf{M}_I + \mathcal{O}_P(N^{-1}) - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{Z}_I \mathbf{K} \right)' (\mathbf{P} \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1)) \\
&= \mathbf{M}_I' \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1) - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{P} \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1) \\
&\quad + \mathcal{O}_P(N^{-1}) - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathcal{O}_P(1) \\
&= \mathbf{M}_I' \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1) - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{P} \mathbf{E} \mathbf{M}_I \\
&\quad - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathcal{O}_P(1) \\
&= \mathbf{M}_I' \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1) - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I \mathbf{P} (\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{E} \mathbf{M}_I \\
&\quad - \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathcal{O}_P(1).
\end{aligned}$$

Earlier, we have already shown that $\mathbf{Z}_I \mathbf{K} = \mathcal{O}_P(T^{-1/2} N^{-1})$ and $(\mathbf{E} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \mathbf{E} \mathbf{M}_I = \mathcal{O}_P(1)$. Since, additionally, $(\mathbf{G} + \kappa_2 \mathbf{I}_{R \times R})^{-1} \leq \mathbf{G} = \mathcal{O}_P(1)$ the above simplifies to

$$\hat{\Lambda}'_I \mathbf{G} \hat{\Lambda}_I = \mathbf{M}'_I \mathbf{E} \mathbf{M}_I + \mathcal{O}_P(1).$$

Consider the column vector $\mathbf{m}_{r,I}$ corresponding to the r th column of the matrix $\mathbf{M}_I$ for $r = 1, \dots, R$. It is then possible to rewrite the diagonal entries of $\mathbf{M}'_I \mathbf{E} \mathbf{M}_I$ as $\sum_{s=1}^R e_s m_{r,s,I}^2$ for $r = 1, \dots, R$. Moreover, using $\mathbf{M}'_I \mathbf{M}_I = \mathbf{I}_{R \times R} - \mathcal{O}_P(N^{-1})$, it follows that $\sum_{s=1}^R e_s m_{r,s,I}^2 \in [e_R + \mathcal{O}_P(1), e_1 + \mathcal{O}_P(1)]$ for all r , and $\widehat{\mathbf{D}} = \mathcal{O}_P(N)$ similar to $\mathbf{E}$. Consequently, this implies that $\widehat{\mathbf{D}}^{-1/2} = \mathcal{O}_P(N^{-1/2})$ (Despois and Doz, 2023, Online Appendix).

Next, define $\widehat{\mathbf{F}} := \widehat{\mathbf{D}}^{-1/2} \hat{\Lambda}'_I \mathbf{X}$, and $\tilde{\mathbf{F}} := \mathbf{E}^{-1/2} \mathbf{P}'_I \mathbf{X}$. Applying Lemma 2 then gives

$$\begin{aligned}
\widehat{\mathbf{F}} &= \widehat{\mathbf{D}}^{-1/2} \left(\mathbf{P} \mathbf{M}_I + \mathcal{O}_P(N^{-1}) - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \right)' \mathbf{X} \\
&= \widehat{\mathbf{D}}^{-1/2} \mathbf{M}'_I \mathbf{P}'_I \mathbf{X}' + \mathcal{O}_P(N^{-1}) + \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} \\
&= \widehat{\mathbf{D}}^{-1/2} \mathbf{M}'_I \mathbf{E}^{1/2} \tilde{\mathbf{F}} + \mathcal{O}_P(N^{-1}) + \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X}.
\end{aligned}$$

Since $\widehat{\mathbf{D}}^{-1/2} \mathbf{M}'_I \mathbf{E}^{1/2} = \mathcal{O}_P(1)$, we define $\mathbf{M} := \widehat{\mathbf{D}}^{-1/2} \mathbf{M}'_I \mathbf{E}^{1/2}$ as in Despois and Doz (2023, Online Appendix).

Inserting (A.11) into $\|\widehat{\mathbf{F}} - \mathbf{M}\tilde{\mathbf{F}}\|_F^2$ then gives

$$\begin{aligned} & \text{tr} \left(\left(\mathcal{O}_P(N^{-1}) - \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} \right)' \left(\mathcal{O}_P(N^{-1}) - \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} \right) \right) \\ &= \text{tr} \left(\mathcal{O}_P(N^{-2}) - \frac{1}{2} \mathcal{O}_P(N^{-1}) \widehat{\mathbf{D}}^{-1/2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} \right. \\ & \quad \left. - \frac{1}{2} \mathcal{O}_P(N^{-1}) \mathbf{X}' (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{-1/2} \right. \\ & \quad \left. + \frac{1}{4} \mathbf{X}' (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{-1/2} \widehat{\mathbf{D}}^{-1/2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} \right). \end{aligned}$$

It is known that $\mathbf{X} = \mathcal{O}_P(T^{1/2}N^{1/2})$, $\widehat{\mathbf{D}}^{-1/2} = \mathcal{O}_P(N^{-1/2})$, and $\mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} = \mathcal{O}_P(T^{-1/2}N^{-1})$. Therefore, the second and third term of the above are

$$\mathcal{O}_P(N^{-1}) \widehat{\mathbf{D}}^{-1/2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} = \mathcal{O}_P(N^{-2}),$$

while the fourth term reads

$$\frac{1}{4} \mathbf{X}' (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{-1} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X} = \mathcal{O}_P(N^{-2}).$$

Using the fact that

$$\left(\mathcal{O}_P(N^{-1}) - \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X}' \right)' \left(\mathcal{O}_P(N^{-1}) - \widehat{\mathbf{D}}^{-1/2} \frac{1}{2} \mathbf{K} \mathbf{Z}'_I (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{X}' \right)$$

is a square matrix, it follows that $\|\widehat{\mathbf{F}}_t - \mathbf{M}\tilde{\mathbf{F}}\|_F^2 = \mathcal{O}_P(N^{-2})$. Consequently, $\|\widehat{\mathbf{F}}_t - \mathbf{M}\tilde{\mathbf{F}}\|_F = \mathcal{O}_P(N^{-1})$.

Lastly, denote $\text{vec}(\mathbf{A}) := (a_{1,1}, \dots, a_{N,1}, a_{1,2}, \dots, a_{N,M})$ for any $\mathbf{A} = (a_{n,m})_{n=1,m=1}^{N,M}$, and define $\boldsymbol{\delta}_1 := \text{vec}(\widehat{\mathbf{F}}_t - \mathbf{M}\tilde{\mathbf{F}})$. From the definition of the matrix maximum norm and the Frobenius norm it is known that $\|\boldsymbol{\delta}_1\|_\infty = \|\widehat{\mathbf{F}}_t - \mathbf{M}\tilde{\mathbf{F}}\|_{\max}$ and $\|\boldsymbol{\delta}_1\|_2 = \|\widehat{\mathbf{F}}_t - \mathbf{M}\tilde{\mathbf{F}}\|_F$. Finally, under the vector norm equivalence, it follows that

$$\|\boldsymbol{\delta}_1\|_\infty \leq \|\boldsymbol{\delta}_1\|_2 = \mathcal{O}_P(N^{-1}). \quad (\text{A.11})$$

■

A.2 Proof of Theorem 2

Proof. Let $\widehat{\boldsymbol{\Lambda}} := \widehat{\boldsymbol{\Lambda}}_I \widehat{\mathbf{D}}^{1/2}$ and define the PCA estimator as $\tilde{\boldsymbol{\Lambda}} := \mathbf{P} \mathbf{E}^{1/2}$ (see, for example, Doz et al., 2011). Using the results of Lemma 2 and defining $\mathbf{M}^{-1} = \mathbf{E}^{-1/2} \mathbf{M}_I^{-1} \widehat{\mathbf{D}}^{1/2}$, the SPCA loadings estimator can be expressed as

$$\begin{aligned} \widehat{\boldsymbol{\Lambda}}_I \widehat{\mathbf{D}}^{1/2} &= \left(\mathbf{P} \mathbf{M}_I + \mathcal{O}_P(N^{-1}) - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \right) \widehat{\mathbf{D}}^{1/2} \\ &= \mathbf{P} \mathbf{E}^{1/2} \mathbf{E}^{-1/2} \mathbf{M}_I \widehat{\mathbf{D}}^{1/2} + \mathcal{O}_P(N^{-1}) \widehat{\mathbf{D}}^{1/2} - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} \\ &= \tilde{\boldsymbol{\Lambda}} \mathbf{M}^{-1} + \mathcal{O}_P(N^{-1/2}) - \frac{1}{2} (\mathbf{G} + \kappa_2 \mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2}, \end{aligned}$$

where the last line follows from the fact that $\widehat{\mathbf{D}}^{1/2} = \mathcal{O}_P(N^{1/2})$. Inserting these results into $\|\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1}\|_F^2$ then gives

$$\begin{aligned}
& \text{tr} \left((\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1})' (\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1}) \right) \\
&= \text{tr} \left(\left(\mathcal{O}_P(N^{-1/2}) - \frac{1}{2}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} \right)' \left(\mathcal{O}_P(N^{-1/2}) - \frac{1}{2}(\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} \right) \right) \\
&= \text{tr} \left(\mathcal{O}_P(N^{-1}) - \frac{1}{2} \widehat{\mathbf{D}}^{1/2} \mathbf{K} \mathbf{Z}_I' (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathcal{O}_P(N^{-1/2}) \right. \\
&\quad \left. - \mathcal{O}_P(N^{-1/2}) \frac{1}{2} (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} \right. \\
&\quad \left. + \frac{1}{4} \widehat{\mathbf{D}}^{1/2} \mathbf{K} \mathbf{Z}_I' (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} \right).
\end{aligned}$$

From [Online Appendix A.1](#), it is known that $\mathbf{K} \mathbf{Z}_I' (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} = \mathcal{O}_P(T^{-1/2}N^{-1})$. This then gives

$$\mathcal{O}_P(N^{-1/2}) \frac{1}{2} (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} = \mathcal{O}_P(T^{-1/2}N^{-1}),$$

and

$$\frac{1}{4} \widehat{\mathbf{D}}^{1/2} \mathbf{K} \mathbf{Z}_I' (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z}_I \mathbf{K} \widehat{\mathbf{D}}^{1/2} = \mathcal{O}_P(T^{-1}N^{-1}). \quad (\text{A.12})$$

Thus, it follows that $(\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1})' (\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1}) = \mathcal{O}_P(N^{-1})$ and, since the matrix is square, its trace is also $\mathcal{O}_P(N^{-1})$. Lastly, defining $\boldsymbol{\delta}_2 := \text{vec}(\widehat{\mathbf{\Lambda}} - \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1})$, just as in [Online Appendix A.1](#), the inequality $\|\boldsymbol{\delta}_2\|_\infty \leq \|\boldsymbol{\delta}_2\|_2 = \mathcal{O}_P(N^{-1/2})$ implies the result. ■

A.3 Proof of Corollary 1

Proof. From [Lemma 2](#) in [Online Appendix A.1](#), it directly follows that

$$\widehat{\mathbf{\Lambda}} = \mathbf{P}\mathbf{M}_\infty + \mathcal{O}_P(N^{-1}) - \frac{1}{2} (\mathbf{G} + \kappa_2\mathbf{I}_N)^{-1} \mathbf{Z} \mathbf{K},$$

where $\mathbf{M}_\infty$ indicates the limit of $\{\mathbf{M}_i\}_{i=0}^\infty$ and $\mathbf{P}\mathbf{M}_\infty$ is the population PCA loadings matrix and identical to an orthogonal transformation of the true loadings matrix $\mathbf{\Lambda}$. We indicate this rotation with the matrix $\mathbf{B}$, such that $\mathbf{P}\mathbf{M}_\infty = \mathbf{B}\mathbf{\Lambda}$. First, we derive the bias assuming that all loadings belong to the active set. Taking expectations, we find that

$$\begin{aligned}
\mathbb{E}(\widehat{\mathbf{\Lambda}}) &= \mathbf{B}\mathbf{\Lambda} - \frac{1}{2} (\bar{\mathbf{G}} + \kappa_2\mathbf{I}_N)^{-1} \mathbb{E}(\mathbf{Z}) \mathbf{K} + o(1) \\
&= \mathbf{B}\mathbf{\Lambda} - \frac{1}{2} (\bar{\mathbf{G}} + \kappa_2\mathbf{I}_N)^{-1} \text{sign}(\mathbf{\Lambda}) \mathbf{K} + o(1),
\end{aligned}$$

where $\bar{\mathbf{G}} = \text{plim } \mathbf{G}$ and $o(1)$ indicates the approximation error in the expectation. Then, the bias of the SPCA loadings matrix is:

$$\mathbb{E}(\hat{\mathbf{\Lambda}} - \mathbf{B}\mathbf{\Lambda}) = -\frac{1}{2}(\bar{\mathbf{G}} + \kappa_2 \mathbf{I}_N)^{-1} \text{sign}(\mathbf{\Lambda}) \mathbf{K} + o(1)$$

Setting $\kappa_{1,1} = \kappa_{2,1} = \dots, \kappa_{r,1} = 0$ and $\kappa_2 = 0$ for the unpenalized PCA estimator, we immediately see that its bias is zero.

We now derive the asymptotic variance of $\hat{\mathbf{\Lambda}}$. The first-order (KKT) condition for the penalized least-squares objective is given by

$$(\mathbf{G} + \kappa_2 \mathbf{I}_N) \hat{\mathbf{\Lambda}} = \mathbf{Z} - \frac{1}{2} \text{sign}(\hat{\mathbf{\Lambda}}) \mathbf{K},$$

and its population analogue by

$$(\bar{\mathbf{G}} + \kappa_2 \mathbf{I}_N) \mathbf{\Lambda} = -\frac{1}{2} \text{sign}(\mathbf{\Lambda}) \mathbf{K}.$$

Subtracting and setting $\mathbf{\Delta}_{\mathbf{\Lambda}} := \hat{\mathbf{\Lambda}} - \mathbf{\Lambda}$ yields

$$\begin{aligned} (\mathbf{G} + \kappa_2 \mathbf{I}_N) (\mathbf{\Lambda} + \mathbf{\Delta}_{\mathbf{\Lambda}}) - (\bar{\mathbf{G}} + \kappa_2 \mathbf{I}_N) \mathbf{\Lambda} &= \mathbf{Z} - \frac{1}{2} (\text{sign}(\hat{\mathbf{\Lambda}}) - \text{sign}(\mathbf{\Lambda})) \mathbf{K} \\ &= \mathbf{Z} - \frac{1}{2} \text{sign}(\mathbf{\Lambda}) \mathbf{K} + o_p(T^{-1/2}). \end{aligned}$$

Dropping higher-order terms, we have

$$(\mathbf{G} + \kappa_2 \mathbf{I}_N) \mathbf{\Delta}_{\mathbf{\Lambda}} = \mathbf{Z} - \frac{1}{2} \text{sign}(\mathbf{\Lambda}) \mathbf{K} + o_p(T^{-1/2}).$$

Defining $\mathbf{H} := (\mathbf{G} + \kappa_2 \mathbf{I}_N)$ and rearranging yields

$$\mathbf{\Delta}_{\mathbf{\Lambda}} = \mathbf{H}^{-1} \mathbf{Z} - \frac{1}{2} \mathbf{H}^{-1} \text{sign}(\mathbf{\Lambda}) \mathbf{K} + o_p(T^{-1/2}).$$

The second term is non-random, hence it holds that

$$\text{Var}(\text{vec}(\hat{\mathbf{\Lambda}})) = \text{Var}(\text{vec}(\mathbf{\Delta}_{\mathbf{\Lambda}})) = \mathbf{H}^{-1} \text{Var}(\text{vec}(\mathbf{Z})) \mathbf{H}^{-1} + o(T^{-1}).$$

Since $\mathbf{Z} = T^{-1} \sum_t \mathbf{x}_t \boldsymbol{\xi}_t'$, it follows from our Assumptions 2 that

$$\sqrt{T} \text{vec}(\mathbf{Z}) \xrightarrow{d} \mathcal{N}(0, \Sigma_{\boldsymbol{\xi}} \otimes \Sigma_{\mathbf{x}}),$$

where $\Sigma_{\boldsymbol{\xi}} := \mathbb{E}(\boldsymbol{\xi}_t \boldsymbol{\xi}_t')$ and $\Sigma_{\mathbf{x}} := \mathbb{E}(\mathbf{x}_t \mathbf{x}_t')$. Finally, we obtain the variance

$$\text{Var}(\text{vec}(\hat{\mathbf{\Lambda}})) = T^{-1} \mathbf{H}^{-1} (\Sigma_{\boldsymbol{\xi}} \otimes \Sigma_{\mathbf{x}}) \mathbf{H}^{-1} + o(T^{-1}).$$

Plugging into the MSE decomposition $\mathbb{E}(\|\hat{\mathbf{\Lambda}}_{\mathcal{A}} - (\mathbf{B}\mathbf{\Lambda})_{\mathcal{A}}\|_F^2) = \|\mathbb{E}(\hat{\mathbf{\Lambda}}_{\mathcal{A}} - (\mathbf{B}\mathbf{\Lambda})_{\mathcal{A}})\|_F^2 + \text{tr}(\text{Var}(\text{vec}(\hat{\mathbf{\Lambda}}_{\mathcal{A}})))$ and subsuming $o(T^{-1})$ into $o(1)$, concludes the proof. ■

A.4 Proof of Proposition 1

Proof. From Theorem 1 and Theorem 2, we know that the estimator of the DFM parameters retrieved via SPCA asymptotically represent an orthogonal transformation of the corresponding PCA estimates $\tilde{\mathbf{f}}_t$ and $\tilde{\mathbf{\Lambda}}$. Therefore, it is possible to express the SPCA estimator for the factors as $\hat{\mathbf{f}}_t = \mathbf{M}\tilde{\mathbf{f}}_t + \mathcal{O}_P(N^{-1})$, for all t , and for the loading matrix as $\hat{\mathbf{\Lambda}} = \tilde{\mathbf{\Lambda}}\mathbf{M}^{-1} + \mathcal{O}_P(N^{-\frac{1}{2}})$. Since $\mathbf{M}$ and $\mathbf{M}^{-1}$ are $\mathcal{O}_P(1)$ (see [Online Appendix A.1](#) and [Online Appendix A.2](#)) this implies that the SPCA estimators have the same asymptotic properties as the PCA estimator outlined in [Doz et al. \(2011\)](#). Since we initialize the KFS similar to [Giannone et al. \(2008\)](#), both $\hat{\mathbf{f}}_t$ and $\hat{\mathbf{\Lambda}}$ converge at least with a rate of $\mathcal{O}_P(N^{-\frac{1}{2}})$ to the PCA estimator, and the PCA estimator converges to an orthogonal transformation of the true factors with $\mathcal{O}_P(N^{-1/2}) + \mathcal{O}_P(T^{-1/2})$, the result follows. ■

B Supplementary Material

B.1 Additional Simulation Results

Table B.1: MSNE reduction in % over 1000 simulated nowcasting exercises using $T = 100$ observations with Hyper-Parameters validated via CV using a non-diagonal variance-covariance matrix for the dense and sparse two-step estimator

		$N = 25$			$N = 50$		
		$R = 1$	$R = 2$	$R = 3$	$R = 1$	$R = 2$	$R = 3$
$s = 0.0$	$\bar{\rho} \approx 0.2$	0.43	-2.16	5.85	4.37	4.36	7.21
	$\bar{\rho} \approx 0.3$	0.02	-1.93	-1.20	1.47	0.36	0.85
	$\bar{\rho} \approx 0.4$	10.38	11.77	12.27	17.04	14.69	15.21
	$\bar{\rho} \approx 0.5$	3.85	8.89	7.85	4.04	10.52	11.46
$s = 0.4$	$\bar{\rho} \approx 0.2$	2.08	1.45	14.57	3.04	3.66	13.32
	$\bar{\rho} \approx 0.3$	1.98	-1.71	-1.71	1.13	-0.67	-0.49
	$\bar{\rho} \approx 0.4$	6.36	18.17	12.65	3.95	19.25	15.77
	$\bar{\rho} \approx 0.5$	1.64	7.03	13.25	4.13	10.13	17.86
$s = 0.8$	$\bar{\rho} \approx 0.2$	4.69	2.45	2.65	5.16	2.90	2.98
	$\bar{\rho} \approx 0.3$	1.56	5.67	1.14	2.58	7.91	0.23
	$\bar{\rho} \approx 0.4$	7.95	8.24	7.49	5.52	3.91	2.39
	$\bar{\rho} \approx 0.5$	8.84	3.79	4.27	3.78	3.11	3.89

Note: $|\bar{\rho}|$ refers to the average absolute cross-correlation coefficients of the measurement errors. s refers to the degree of sparsity which indicates the proportion of zero elements in each column of the factor loading matrix $\mathbf{\Lambda}$. The colour coding indicates the relative performance of the sparse models versus the benchmark per number of factors, with green indicating better and red indicating worse performance. The gradient of the colour indicates the magnitude of the MSNE improvement/deterioration.

Table B.2: MSNE reduction in % over 1000 simulated nowcasting exercises using $T = 100$ observations with Hyper-Parameters validated via the BIC using a non-diagonal variance-covariance matrix for the dense and sparse two-step estimator

		$N = 25$			$N = 50$		
		$R = 1$	$R = 2$	$R = 3$	$R = 1$	$R = 2$	$R = 3$
$s = 0.0$	$\bar{\rho} \approx 0.2$	-36.11	-14.80	-2.84	-21.31	-15.74	-4.00
	$\bar{\rho} \approx 0.3$	-18.09	-11.10	-9.17	-14.98	-11.26	-8.05
	$\bar{\rho} \approx 0.4$	-21.41	-8.90	-1.20	-10.93	-0.36	-0.20
	$\bar{\rho} \approx 0.5$	-22.75	-4.29	-1.04	-13.84	-3.51	-1.46
$s = 0.4$	$\bar{\rho} \approx 0.2$	2.12	-12.13	-8.90	1.21	-13.00	-14.28
	$\bar{\rho} \approx 0.3$	1.05	-12.11	-24.40	1.45	-6.97	-15.91
	$\bar{\rho} \approx 0.4$	-2.41	-9.70	-4.20	-0.68	-0.73	-0.66
	$\bar{\rho} \approx 0.5$	-0.21	-8.43	-1.46	0.43	-2.93	-0.84
$s = 0.8$	$\bar{\rho} \approx 0.2$	2.64	-2.47	-3.06	1.09	-4.59	-2.17
	$\bar{\rho} \approx 0.3$	0.74	-5.60	-3.57	0.26	-10.46	-2.77
	$\bar{\rho} \approx 0.4$	1.06	-1.44	-2.67	0.19	-0.42	-0.05
	$\bar{\rho} \approx 0.5$	1.49	-4.87	-4.37	-0.03	-0.69	-0.79

Note: $|\bar{\rho}|$ refers to the average absolute cross-correlation coefficients of the measurement errors. s refers to the degree of sparsity which indicates the proportion of zero elements in each column of the factor loading matrix $\mathbf{\Lambda}$. The colour coding indicates the relative performance of the sparse models versus the benchmark per number of factors, with green indicating better and red indicating worse performance. The gradient of the colour indicates the magnitude of the MSNE improvement/deterioration.

Table B.3: MSNE reduction in % over 1000 simulated nowcasting exercises using $T = 100$ observations with Hyper-Parameters validated via the BIC using a non-diagonal variance-covariance matrix for the sparse and a diagonal variance covariance matrix for the dense two-step estimator

		$N = 25$			$N = 50$		
		$R = 1$	$R = 2$	$R = 3$	$R = 1$	$R = 2$	$R = 3$
$s = 0.0$	$\bar{\rho} \approx 0.2$	0.80	-1.17	-0.20	-4.99	1.05	4.92
	$\bar{\rho} \approx 0.3$	2.14	0.94	0.58	0.93	3.33	0.71
	$\bar{\rho} \approx 0.4$	1.57	-2.56	4.17	3.43	1.22	0.22
	$\bar{\rho} \approx 0.5$	-1.56	3.83	4.28	1.09	3.02	1.70
$s = 0.4$	$\bar{\rho} \approx 0.2$	0.18	6.30	1.29	0.34	12.77	-0.88
	$\bar{\rho} \approx 0.3$	0.16	2.97	4.23	0.03	9.53	1.93
	$\bar{\rho} \approx 0.4$	11.33	9.65	1.74	2.08	1.66	2.56
	$\bar{\rho} \approx 0.5$	1.84	10.39	-1.28	-0.04	2.08	1.50
$s = 0.8$	$\bar{\rho} \approx 0.2$	6.03	0.71	-2.84	9.63	-0.66	-2.21
	$\bar{\rho} \approx 0.3$	0.50	0.87	-1.58	1.40	4.40	-1.47
	$\bar{\rho} \approx 0.4$	16.53	5.33	0.08	-0.65	1.69	2.82
	$\bar{\rho} \approx 0.5$	8.32	-1.41	-2.90	0.43	1.46	1.26

Note: $|\bar{\rho}|$ refers to the average absolute cross-correlation coefficients of the measurement errors. s refers to the degree of sparsity which indicates the proportion of zero elements in each column of the factor loading matrix $\mathbf{\Lambda}$. The colour coding indicates the relative performance of the sparse models versus the benchmark per number of factors, with green indicating better and red indicating worse performance. The gradient of the colour indicates the magnitude of the MSNE improvement/deterioration.

B.2 Additional Empirical Results

Table B.4: Relative nowcasting results with statistical tests disaggregated by subperiods with Giacomini-White-Type Regression using loss differential lags

Time Period	First Month of Quarter			Second Month of Quarter			Third Month of Quarter		
	DFM	DFM EM	SDFM EM	DFM	DFM EM	SDFM EM	DFM	DFM EM	SDFM EM
Relative Improvement (%)									
Full Sample w/o Covid	21.82	17.30	11.49	17.52	20.76	13.15	6.65	7.14	4.98
Pre-Covid	22.65	17.22	12.11	20.13	24.62	13.58	8.11	9.58	1.65
Covid	38.95	-136.85	-129.61	38.31	-122.98	-93.96	33.35	-130.98	-102.81
Post-Covid	18.10	17.64	8.74	8.04	5.53	11.73	2.70	0.30	12.53
Dot-Com Crisis	4.78	36.72	39.75	14.57	45.90	48.70	10.20	38.17	45.35
Financial Crisis	76.87	65.76	50.20	83.59	85.97	65.60	35.16	37.59	-17.47
No Crisis	-8.32	-2.73	0.18	0.42	0.14	2.94	0.34	-1.79	-0.45
Giacomini-White-Type Regression									
No Crisis	-0.022 (0.025)	-0.002 (0.017)	0.012 (0.015)	0.013 (0.019)	0.010 (0.012)	0.017 (0.012)	0.008 (0.011)	-0.001 (0.013)	0.002 (0.015)
Covid	28.239*** (7.857)	-26.330* (15.252)	-25.795** (10.070)	25.726*** (2.666)	-24.375* (14.042)	-21.486** (9.006)	15.653*** (2.047)	-19.492*** (2.624)	-18.560*** (3.687)
Post-Covid	0.171** (0.070)	0.149 (0.149)	0.064* (0.033)	0.172* (0.097)	0.057 (0.059)	0.089 (0.061)	0.086*** (0.032)	0.015 (0.026)	0.089* (0.052)
Dot-Com Crisis	-0.049 (0.043)	0.422*** (0.132)	0.476*** (0.096)	0.024*** (0.008)	0.411*** (0.110)	0.530*** (0.097)	0.016** (0.006)	0.214*** (0.017)	0.316*** (0.036)
Financial Crisis	2.377* (1.362)	1.341 (0.889)	0.635*** (0.234)	1.211*** (0.283)	1.580* (0.877)	0.465*** (0.158)	0.211*** (0.033)	0.270*** (0.066)	-0.123** (0.053)
ΔL_{t-1}	-0.439*** (0.163)	-0.540** (0.252)	-0.562*** (0.164)	-0.419*** (0.060)	-0.539** (0.251)	-0.603*** (0.148)	-0.190*** (0.053)	-0.296*** (0.046)	-0.442*** (0.059)
ΔL_{t-2}	-0.510*** (0.142)	-0.444 (0.269)	-0.430** (0.176)	-0.482*** (0.052)	-0.445* (0.267)	-0.392** (0.163)	-0.333*** (0.042)	-0.347*** (0.044)	-0.311*** (0.068)

*: $p < 0.10$, **: $p < 0.05$, ***: $p < 0.01$

The crisis periods are constructed according to [Hamilton \(2026\)](#) as: 2020Q1 to 2021Q1 for “Covid”, 2001Q1 to 2001Q4 for “Dotcom Crisis”, 2007Q4 to 2009Q2 for “Financial Crisis”. “Full Sample w/o Covid” refers to the entire out-of-sample period from 1999Q3 to 2025Q3 excluding the covid period from 2020Q1 to 2021Q1. “Pre-Covid” refers to the period of 1999Q3 to 2019Q4. “Post-Covid” refers to the period from 2021Q2 to 2025Q3. All periods outside of the periods of “Covid”, “Post-Covid”, “Dotcom Crisis”, and “Financial Crisis” are referred to as “No Crisis”. The MSNE Reduction is computed as in ??.

“DFM” denotes the dense two-step estimator by [Giannone et al. \(2008\)](#). “DFM EM” denotes the dense expectation-maximisation estimator by [Bańbura and Modugno \(2014\)](#). “SDFM EM” denotes the sparse expectation-maximisation estimator by [Mosley et al. \(2023\)](#) with hyper-parameters validated via the BIC.

ΔL_{t-1} and ΔL_{t-2} denote the first and second lag of the loss differential between the two models, i.e., $\Delta L_t := L_{t,\text{SDFM}} - L_{t,i}$, where, for all t , $L_{t,i} = (\hat{x}_{t,i} - x_{t,i})^2$ for nowcast $\hat{x}_{t,i}$ retrieved via model $i \in \{\text{DFM}, \text{DFM EM}, \text{SDFM EM}\}$ of observation x_t . The individual coefficient significant tests where conducted using heteroscedasticity- and autocorrelation-robust Newey-West type variance covariance matrix estimators.

Table B.5: Mean squared nowcasting error disaggregated by subperiods

Time Period	SDFM (CV)	SDFM (BIC)	DFM	DFM EM	SDFM EM
First Month of Quarter					
Full Sample w/o Covid	0.269	0.288	0.344	0.325	0.304
Pre-Covid	0.263	0.269	0.340	0.317	0.299
Covid	22.868	21.243	37.456	9.655	9.959
Post-Covid	0.298	0.378	0.364	0.362	0.327
Dotcom Crisis	0.308	0.346	0.324	0.488	0.512
Financial Crisis	0.384	0.602	1.662	1.123	0.772
No-Crisis	0.251	0.239	0.231	0.244	0.251
Second Month of Quarter					
Full Sample w/o Covid	0.230	0.258	0.279	0.290	0.265
Pre-Covid	0.211	0.230	0.264	0.280	0.244
Covid	22.185	23.939	35.964	9.949	11.438
Post-Covid	0.322	0.393	0.350	0.341	0.365
Dotcom Crisis	0.239	0.262	0.280	0.442	0.466
Financial	0.136	0.350	0.831	0.972	0.396
No Crisis	0.216	0.218	0.217	0.216	0.222
Last Month of Quarter					
Full Sample w/o Covid	0.208	0.229	0.223	0.225	0.219
Pre-Covid	0.181	0.192	0.197	0.200	0.184
Covid	20.863	20.816	31.300	9.032	10.287
Post-Covid	0.342	0.406	0.352	0.343	0.391
Dotcom Crisis	0.188	0.192	0.210	0.305	0.345
Financial Crisis	0.312	0.427	0.481	0.499	0.265
No Crisis	0.167	0.169	0.168	0.165	0.167

The crisis periods are constructed according to [Hamilton \(2026\)](#) as: 2020Q1 to 2021Q1 for “Covid”, 2001Q1 to 2001Q4 for “Dotcom Crisis”, 2007Q4 to 2009Q2 for “Financial Crisis”. “Full Sample w/o Covid” refers to the entire out-of-sample period from 1999Q3 to 2025Q3 excluding the covid period from 2020Q1 to 2021Q1. “Pre-Covid” refers to the period of 1999Q3 to 2019Q4. “Post-Covid” refers to the period from 2021Q2 to 2025Q3. All periods outside of the periods of “Covid”, “Post-Covid”, “Dotcom Crisis”, and “Financial Crisis” are referred to as “No Crisis”.

“SDFM (CV)” denotes the two-step estimator for SDFMs proposed in this study with hyper parameters selected via cross-validation. “SDFM (BIC)” denotes the two-step estimator for SDFMs proposed in this study with hyper parameters selected via the BIC. “DFM” denotes the two-step estimator for DFMs by [Giannone et al. \(2008\)](#). “DFM EM” denotes the expectation-maximisation estimator for DFMs by [Bańbura and Modugno \(2014\)](#). “SDFM EM” denotes the expectation-maximisation estimator for SDFMs by [Mosley et al. \(2023\)](#) with hyper-parameters validated via the BIC.

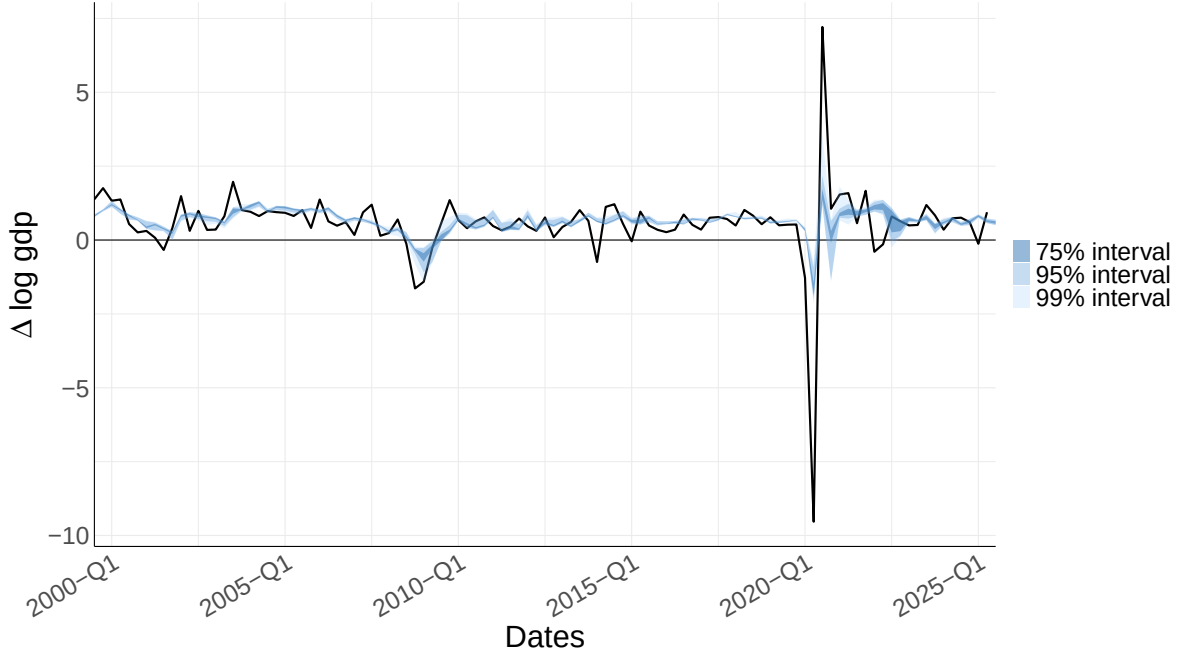
Table B.6: Mean squared one-step-ahead forecasting error disaggregated by subperiods

Time Period	SDFM (CV)	SDFM (BIC)	DFM	DFM EM	SDFM EM
First Month of Quarter					
Full Sample w/o Covid	1.393	1.375	1.388	1.395	1.387
Pre-Covid	1.611	1.588	1.611	1.612	1.604
Covid	21.189	25.874	18.222	52.635	61.839
Post-Covid	0.278	0.283	0.246	0.285	0.275
Dotcom Crisis	1.147	0.824	0.581	0.999	0.833
Financial Crisis	1.400	1.404	1.775	1.654	1.564
No-Crisis	1.647	1.635	1.640	1.634	1.639
Second Month of Quarter					
Full Sample w/o Covid	1.388	1.379	1.414	1.408	1.392
Pre-Covid	1.603	1.593	1.643	1.627	1.610
Covid	19.819	23.001	19.017	46.348	57.104
Post-Covid	0.283	0.287	0.244	0.286	0.276
Dotcom Crisis	0.868	0.698	0.589	0.940	0.860
Financial Crisis	1.533	1.523	1.935	1.826	1.577
No-Crisis	1.639	1.635	1.662	1.639	1.643
Last Month of Quarter					
Full Sample w/o Covid	0.332	0.330	0.313	0.346	0.340
Pre-Covid	0.342	0.338	0.326	0.359	0.353
Covid	35.051	37.089	34.326	55.113	62.710
Post-Covid	0.282	0.286	0.244	0.281	0.272
Dotcom Crisis	0.785	0.574	0.457	0.843	0.927
Financial Crisis	1.055	1.135	1.154	1.278	1.163
No-Crisis	0.247	0.247	0.237	0.241	0.241

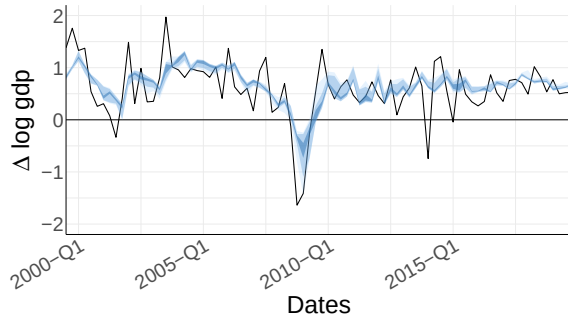
The crisis periods are constructed according to [Hamilton \(2026\)](#) as: 2020Q1 to 2021Q1 for “Covid”, 2001Q1 to 2001Q4 for “Dotcom Crisis”, 2007Q4 to 2009Q2 for “Financial Crisis”. “Full Sample w/o Covid” refers to the entire out-of-sample period from 1999Q3 to 2025Q3 excluding the covid period from 2020Q1 to 2021Q1. “Pre-Covid” refers to the period of 1999Q3 to 2019Q4. “Post-Covid” refers to the period from 2021Q2 to 2025Q3. All periods outside of the periods of “Covid”, “Post-Covid”, “Dotcom Crisis”, and “Financial Crisis” are referred to as “No Crisis”.

“SDFM (CV)” denotes the two-step estimator for SDFMs proposed in this study with hyper parameters selected via cross-validation. “SDFM (BIC)” denotes the two-step estimator for SDFMs proposed in this study with hyper parameters selected via the BIC. “DFM” denotes the two-step estimator for DFMs by [Giannone et al. \(2008\)](#). “DFM EM” denotes the expectation-maximisation estimator for DFMs by [Bańbura and Modugno \(2014\)](#). “SDFM EM” denotes the expectation-maximisation estimator for SDFMs by [Mosley et al. \(2023\)](#) with hyper-parameters validated via the BIC.

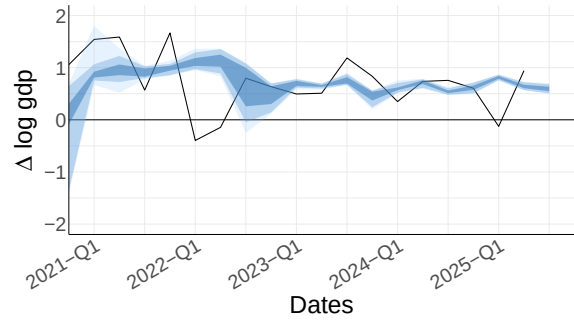
Figure B.1: Intervals of the distribution of nowcasts across posterior hyper-parameter draws compared to the realisations of US GDP growth from 1999Q3 to 2025Q3



(a) Full Sample



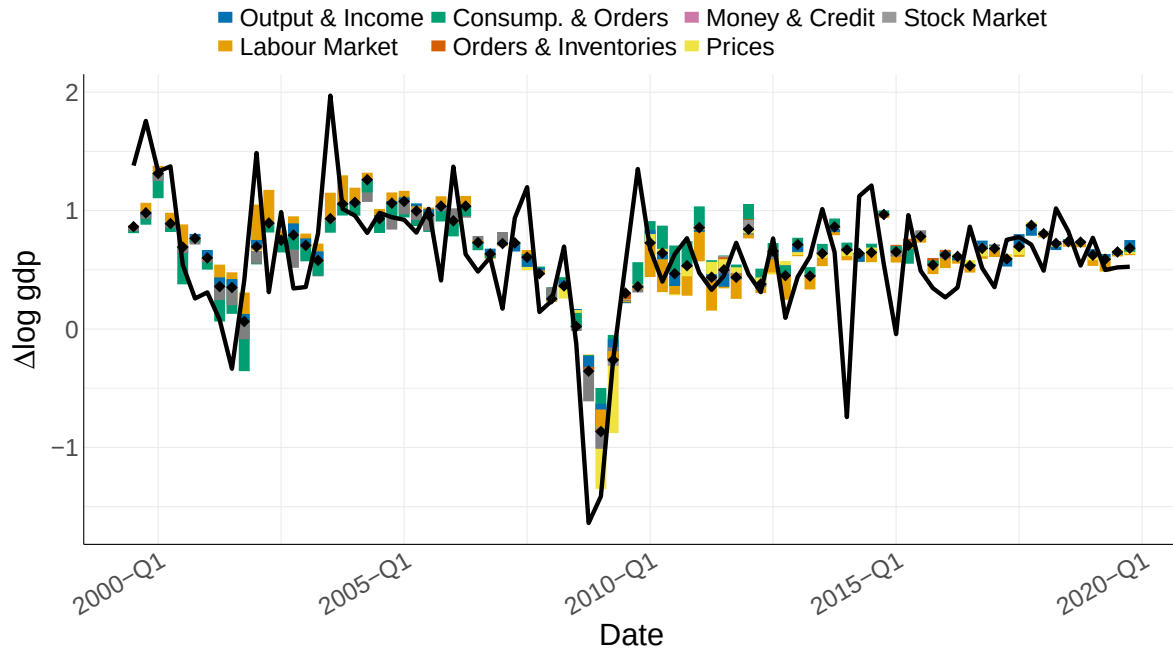
(b) Pre-Covid



(c) Post-Covid

“Full Sample” refers to the entire out-of-sample period from 1999Q3 to 2025Q3. “Pre-Covid” refers to the period of 1999Q3 to 2019Q4. “Post-Covid” refers to the calculated nowcasts for the period from 2021Q2 to 2025Q3. The black line represent the realisations of the first log differences of the US GDP as reported in the respective vintage. The ribbon bands depict the 75-, 95-, and 99-percentiles of the distribution of nowcasts obtained by providing a random set of hyper-parameters. For the construction of the bands, 10,000 nowcasts were computed at each date using a set of random hyperparameters.

Figure B.2: Group omission impact on the full model nowcast compared to the realisations over the pre-Covid period from 1999Q3 to 2019Q4



The coloured bars indicate the relative difference between the full model nowcast (depicted as a black diamond) and the nowcast obtained by omitting the respective variables belonging to the given group.

Table B.7: Impact of omitting variable groups on the full model nowcast against realisations outcomes from 1999Q3 to 2025Q2

		Output		Consump.	Orders	Money	Interest		Stock	
	Full	&	Labour	&	&	&	&			
Date	Model	Income	Market	Orders	Inventories	Credit	FX	Prices	Market	Realisation
Sept 1999	0.864	0.856	0.876	0.809	0.859	0.865	0.825	0.862	0.867	1.380
Dec 1999	0.981	0.964	1.066	0.881	0.981	0.988	0.930	0.984	0.981	1.757
Mar 2000	1.314	1.313	1.377	1.104	1.311	1.313	1.250	1.311	1.317	1.330
Jun 2000	0.890	0.884	0.982	0.819	0.892	0.889	0.855	0.892	0.888	1.373
Sept 2000	0.692	0.703	0.883	0.377	0.696	0.692	0.642	0.692	0.686	0.542
Dec 2000	0.765	0.803	0.750	0.735	0.770	0.764	0.714	0.765	0.752	0.258
Mar 2001	0.600	0.668	0.587	0.502	0.603	0.605	0.563	0.596	0.583	0.309
Jun 2001	0.358	0.436	0.544	0.063	0.360	0.353	0.244	0.356	0.345	0.077
Sept 2001	0.350	0.422	0.479	0.128	0.352	0.352	0.201	0.352	0.338	-0.336
Dec 2001	0.064	0.127	0.308	-0.356	0.054	0.059	-0.086	0.067	0.056	0.409
Mar 2002	0.695	0.752	1.051	0.546	0.671	0.676	0.552	0.659	0.697	1.486
Jun 2002	0.895	0.894	1.175	0.815	0.894	0.892	0.882	0.895	0.879	0.311
Sept 2002	0.750	0.741	0.832	0.647	0.750	0.754	0.740	0.750	0.733	0.987
Dec 2002	0.794	0.892	0.952	0.671	0.802	0.787	0.517	0.794	0.806	0.343
Mar 2003	0.706	0.765	0.806	0.572	0.714	0.698	0.688	0.703	0.704	0.354
Jun 2003	0.579	0.656	0.721	0.450	0.586	0.585	0.445	0.584	0.582	0.807
Sept 2003	0.932	0.932	1.151	0.813	0.921	0.926	0.903	0.928	0.928	1.971
Dec 2003	1.057	1.050	1.298	0.959	1.050	1.047	1.061	1.060	1.055	1.015
Mar 2004	1.069	1.047	1.193	0.957	1.071	1.073	1.068	1.066	1.067	0.958
Jun 2004	1.261	1.262	1.320	1.154	1.261	1.264	1.073	1.261	1.257	0.812
Sept 2004	0.931	0.962	1.014	0.811	0.936	0.925	0.932	0.935	0.934	0.981
Dec 2004	1.063	1.081	1.153	0.949	1.066	1.063	0.839	1.062	1.067	0.944
Mar 2005	1.078	1.112	1.167	0.942	1.084	1.076	0.969	1.077	1.077	0.923
Jun 2005	0.997	1.061	1.037	0.870	0.998	0.993	0.921	0.999	0.997	0.814
Sept 2005	0.962	1.012	1.025	0.820	0.958	0.965	0.829	0.956	0.967	1.015
Dec 2005	1.038	1.026	1.121	0.908	1.034	1.042	0.997	1.043	1.038	0.410

Table B.7: Impact of omitting variable groups on the full model nowcast against realisations from 1999Q3 to 2025Q2
(continued)

		Output		Consump.	Orders	Money	Interest Rates		Stock	
	Full	&	Labour	&	&	&	&		Market	Realisation
Date	Model	Income	Market	Orders	Inventories	Credit	FX	Prices		
Mar 2006	0.914	0.910	0.896	0.782	0.908	0.916	1.019	0.912	0.917	1.371
Jun 2006	1.038	1.027	1.124	0.954	1.039	1.034	0.940	1.038	1.038	0.632
Sept 2006	0.731	0.718	0.752	0.665	0.730	0.734	0.786	0.732	0.733	0.485
Dec 2006	0.634	0.679	0.592	0.601	0.634	0.639	0.615	0.634	0.637	0.605
Mar 2007	0.722	0.734	0.699	0.710	0.722	0.719	0.819	0.721	0.718	0.172
Jun 2007	0.728	0.654	0.726	0.713	0.720	0.727	0.731	0.733	0.728	0.938
Sept 2007	0.606	0.524	0.667	0.633	0.596	0.601	0.590	0.495	0.604	1.198
Dec 2007	0.468	0.516	0.494	0.527	0.459	0.466	0.500	0.466	0.470	0.144
Mar 2008	0.255	0.231	0.231	0.353	0.250	0.260	0.351	0.256	0.247	0.239
Jun 2008	0.364	0.411	0.358	0.438	0.351	0.374	0.354	0.258	0.364	0.697
Sept 2008	0.022	0.169	0.018	0.136	0.027	0.005	-0.017	0.160	0.021	-0.128
Dec 2008	-0.356	-0.227	-0.216	-0.221	-0.320	-0.342	-0.611	-0.350	-0.366	-1.638
Mar 2009	-0.866	-0.631	-0.679	-0.498	-0.844	-0.867	-1.011	-1.350	-0.830	-1.412
Jun 2009	-0.262	-0.087	-0.187	-0.050	-0.249	-0.262	-0.153	-0.879	-0.311	-0.255
Sept 2009	0.303	0.218	0.226	0.296	0.253	0.308	0.237	0.250	0.248	0.553
Dec 2009	0.358	0.346	0.335	0.564	0.357	0.361	0.309	0.376	0.359	1.351
Mar 2010	0.728	0.835	0.439	0.911	0.729	0.723	0.728	0.803	0.730	0.675
Jun 2010	0.641	0.589	0.311	0.873	0.654	0.636	0.634	0.677	0.651	0.400
Sept 2010	0.468	0.364	0.294	0.705	0.466	0.472	0.467	0.285	0.470	0.632
Dec 2010	0.535	0.529	0.281	0.742	0.521	0.539	0.529	0.445	0.523	0.767
Mar 2011	0.856	0.820	0.572	1.036	0.838	0.865	0.863	0.840	0.844	0.474
Jun 2011	0.437	0.463	0.155	0.581	0.444	0.429	0.404	0.568	0.442	0.331
Sept 2011	0.500	0.357	0.343	0.621	0.605	0.498	0.469	0.590	0.618	0.450
Dec 2011	0.437	0.420	0.255	0.545	0.467	0.434	0.435	0.521	0.474	0.728
Mar 2012	0.842	0.800	0.765	1.056	0.927	0.846	0.857	0.807	0.921	0.464
Jun 2012	0.381	0.359	0.304	0.510	0.371	0.378	0.368	0.435	0.369	0.311
Sept 2012	0.658	0.611	0.478	0.726	0.646	0.665	0.588	0.460	0.649	0.765

Table B.7: Impact of omitting variable groups on the full model nowcast against realisations from 1999Q3 to 2025Q2
(continued)

		Output		Consump.	Orders	Money	Interest Rates		Stock	
	Full	&	Labour	&	&	&	&	Prices	Market	Realisation
Date	Model	Income	Market	Orders	Inventories	Credit	FX			
Dec 2012	0.451	0.538	0.247	0.534	0.460	0.449	0.459	0.574	0.458	0.094
Mar 2013	0.713	0.652	0.615	0.771	0.707	0.720	0.649	0.625	0.702	0.440
Jun 2013	0.447	0.464	0.334	0.523	0.455	0.444	0.453	0.459	0.438	0.613
Sept 2013	0.637	0.652	0.531	0.719	0.644	0.650	0.665	0.601	0.644	1.013
Dec 2013	0.863	0.819	0.792	0.933	0.867	0.862	0.861	0.844	0.855	0.648
Mar 2014	0.668	0.657	0.580	0.730	0.674	0.669	0.685	0.616	0.665	-0.744
Jun 2014	0.639	0.567	0.566	0.641	0.629	0.637	0.642	0.627	0.631	1.122
Sept 2014	0.647	0.651	0.564	0.720	0.657	0.645	0.644	0.688	0.644	1.212
Dec 2014	0.967	0.961	0.937	1.002	0.951	0.967	0.963	0.969	0.967	0.548
Mar 2015	0.653	0.700	0.561	0.700	0.711	0.653	0.612	0.657	0.653	-0.043
Jun 2015	0.706	0.706	0.679	0.551	0.762	0.706	0.734	0.710	0.707	0.962
Sept 2015	0.781	0.787	0.730	0.774	0.781	0.781	0.834	0.782	0.780	0.491
Dec 2015	0.540	0.600	0.464	0.577	0.599	0.540	0.536	0.543	0.540	0.344
Mar 2016	0.625	0.651	0.514	0.644	0.668	0.625	0.596	0.627	0.626	0.267
Jun 2016	0.613	0.579	0.552	0.619	0.605	0.616	0.613	0.632	0.605	0.351
Sept 2016	0.534	0.549	0.476	0.547	0.537	0.534	0.540	0.582	0.531	0.864
Dec 2016	0.681	0.747	0.592	0.700	0.678	0.680	0.712	0.615	0.680	0.515
Mar 2017	0.680	0.706	0.608	0.684	0.673	0.686	0.714	0.701	0.674	0.353
Jun 2017	0.592	0.527	0.601	0.596	0.596	0.588	0.587	0.633	0.589	0.754
Sept 2017	0.696	0.801	0.609	0.699	0.690	0.702	0.682	0.626	0.689	0.777
Dec 2017	0.875	0.787	0.882	0.886	0.865	0.870	0.880	0.903	0.871	0.712
Mar 2018	0.804	0.828	0.772	0.795	0.804	0.802	0.799	0.825	0.800	0.493
Jun 2018	0.722	0.667	0.705	0.712	0.719	0.720	0.743	0.696	0.725	1.019
Sept 2018	0.737	0.740	0.695	0.759	0.738	0.737	0.749	0.737	0.734	0.825
Dec 2018	0.734	0.735	0.691	0.745	0.736	0.730	0.736	0.727	0.743	0.536
Mar 2019	0.623	0.674	0.533	0.622	0.631	0.627	0.626	0.581	0.625	0.770
Jun 2019	0.589	0.633	0.485	0.595	0.588	0.587	0.585	0.605	0.585	0.498

Table B.7: Impact of omitting variable groups on the full model nowcast against realisations from 1999Q3 to 2025Q2
(continued)

		Output		Consump.	Orders	Money	Interest Rates		Stock	
	Full	&	Labour	&	&	&	&			
Date	Model	Income	Market	Orders	Inventories	Credit	FX	Prices	Market	Realisation
Sept 2019	0.651	0.627	0.613	0.643	0.651	0.654	0.636	0.667	0.650	0.520
Dec 2019	0.683	0.751	0.623	0.652	0.680	0.693	0.684	0.633	0.678	0.526
Mar 2020	0.371	0.516	0.265	0.314	0.386	0.376	0.360	0.496	0.375	-1.281
Jun 2020	-1.319	-1.019	-1.168	-1.604	-1.172	-1.252	-1.442	-2.015	-1.234	-9.533
Sept 2020	1.414	1.868	1.146	2.004	1.618	1.337	1.405	2.391	1.406	7.212
Dec 2020	0.723	-1.271	0.645	-1.013	0.573	0.677	0.717	-1.526	0.693	1.059
Mar 2021	0.926	0.819	0.822	1.040	0.840	0.932	0.928	0.823	0.907	1.541
Jun 2021	0.911	0.981	0.792	1.183	0.825	0.888	0.913	1.010	0.896	1.588
Sept 2021	0.769	0.829	0.666	0.833	0.823	0.774	0.766	0.767	0.768	0.569
Dec 2021	1.039	0.951	0.841	0.946	1.018	1.039	1.057	0.898	1.044	1.667
Mar 2022	1.066	1.040	0.965	1.173	1.044	1.054	1.093	0.938	1.078	-0.396
Jun 2022	0.841	0.552	0.763	0.882	0.936	0.848	0.880	0.737	0.866	-0.145
Sept 2022	0.283	0.253	0.356	-0.086	0.445	0.307	0.283	0.608	0.307	0.798
Dec 2022	0.517	0.592	0.375	0.159	0.532	0.518	0.555	0.596	0.527	0.635
Mar 2023	0.703	0.759	0.549	0.735	0.687	0.699	0.697	0.832	0.698	0.496
Jun 2023	0.611	0.553	0.600	0.630	0.639	0.613	0.609	0.638	0.610	0.510
Sept 2023	0.750	0.763	0.708	0.827	0.705	0.745	0.777	0.670	0.740	1.187
Dec 2023	0.368	0.539	0.218	0.298	0.412	0.380	0.360	0.486	0.375	0.835
Mar 2024	0.750	0.805	0.606	0.695	0.719	0.743	0.748	0.748	0.733	0.350
Jun 2024	0.609	0.530	0.666	0.684	0.648	0.612	0.612	0.632	0.608	0.736
Sept 2024	0.625	0.698	0.547	0.562	0.605	0.623	0.608	0.697	0.619	0.757
Dec 2024	0.570	0.674	0.482	0.684	0.568	0.571	0.590	0.468	0.563	0.605
Mar 2025	0.779	0.669	0.849	0.797	0.812	0.784	0.787	0.750	0.782	-0.126
Jun 2025	0.700	0.558	0.702	0.678	0.636	0.702	0.670	0.658	0.712	0.942

For each column, except “Full Model”, the values correspond to the nowcast obtained after excluding all variables belonging to the respective group. The nowcasts are computed at the end of the final month of the respective quarter. The “Realisation” column reports the realised value of GDP growth.

C Algorithms

In the upcoming section, the following conventions are used: For any two vectors $\mathbf{a} := (a_n)_{n=1}^N$ and $\mathbf{b} := (b_m)_{m=1}^M$, we define the concatenation operator $\oplus$ such that $\mathbf{a} \oplus \mathbf{b} := (a_1, \dots, a_N, b_1, \dots, b_M)'$ and $\mathbf{a} \oplus c := (a_1, \dots, a_N, c)'$. Let $\mathcal{I}$ be a sequence of integer values and $\mathbf{A} = (a_{n,m})_{n=1,m=1}^{N,M}$ be a matrix. Assuming $\min \mathcal{I} \geq 1$ and $\max \mathcal{I} \leq N$, $\mathbf{A}[\mathcal{I}, \cdot]$ refers to the submatrix of $\mathbf{A}$ that is constructed by concatenating the rows with index corresponding to the integers in $\mathcal{I}$ in order. Similarly, for any vector $\mathbf{a} := (a_n)_{n=1}^N$ and integer sequence $\mathcal{I}$ with $\min \mathcal{I} \geq 1$ and $\max \mathcal{I} \leq N$, $\mathbf{a}[\mathcal{I}]$ denotes the subvector consisting of the elements of $\mathbf{a}$ with corresponding index $i \in \mathcal{I}$ in order. $\mathbf{a}[n_1 : n_m]$ refers to the subvector consisting of the elements with index $1 \leq n_1 \leq n_m$ to index $n_1 \leq n_m \leq N$. Initialising an empty vector is denoted as $\mathbf{a} = (\cdot)$. The operation of removing an element a_n from vector $\mathbf{a}$ is defined as $\mathbf{a} \ominus a_n := (a_1, \dots, a_{n-1}, a_{n+1}, \dots, a_N)$. For two vectors $\mathbf{a}, \mathbf{b}$ of equal length, the element-wise division is defined as $\mathbf{a} \oslash \mathbf{b} := (a_1/b_1, \dots, a_N/b_N)$. For any boolean expression $d \in \{\text{true}, \text{false}\}$, $\neg d$ represents the negation.

Algorithm 1 Sparse Principal Components Analysis (based on [Zou and Hastie, 2020](#))

```

1: input data matrix  $\mathbf{X}$ 
2: input size constraints  $\kappa_2, \kappa_{1,1} \dots, \kappa_{r,1}$ 
3: input conversion threshold  $\epsilon$ 
4: compute  $\mathbf{X}' = \mathbf{U} \mathbf{D} \mathbf{V}'$ 
5:  $\mathbf{A} \leftarrow (\mathbf{v}_1, \dots, \mathbf{v}_R)$ , where  $\mathbf{v}_1, \dots, \mathbf{v}_R$  correspond to the first  $R$  columns of  $\mathbf{V}$ .
6:  $\tilde{\mathbf{A}} \leftarrow (\infty)_{t=1,r=1}^{T,R}$ 
7: while  $\|\mathbf{A} - \tilde{\mathbf{A}}\|_F > \epsilon$  do
8:    $\tilde{\mathbf{A}} \leftarrow \mathbf{A}$ 
9:   for  $r \in \{1, \dots, R\}$  do
10:    Use Algorithm 2 to solve
        
$$\hat{\lambda}_r \leftarrow \underset{\lambda_r}{\operatorname{argmin}} \left\{ (\boldsymbol{\alpha}_r - \lambda_r)' \mathbf{X}_r \mathbf{X}_r' (\boldsymbol{\alpha}_r - \lambda_r) + \kappa_2 \|\lambda_r\|_2 + \kappa_{r,1} \|\lambda_r\|_1 \right\}$$

11:   end for
12:    $\hat{\Lambda} \leftarrow (\hat{\lambda}_1, \dots, \hat{\lambda}_R)$ 
13:    $\mathbf{B} \leftarrow \mathbf{X} \mathbf{X}' \hat{\Lambda}$ 
14:   compute  $\mathbf{B} = \mathbf{U} \mathbf{D} \mathbf{V}'$ .
15:    $\mathbf{A} \leftarrow \mathbf{U} \mathbf{V}'$ 
16: end while
17: return  $\hat{\Lambda}$ 

```

Algorithm 2 Least Angle Regression (based on [Efron et al., 2004](#); [Zou and Hastie, 2005, 2020](#))

```

1: input  $\boldsymbol{\alpha}_r$  from Algorithm 1

```

```

2: input data matrix  $\mathbf{X}$ 
3: input size constraints  $\kappa_2, \kappa_{1,1} \dots, \kappa_{r,1}$ 
4: input alternative stopping criterion  $M$  (number of non-zero elements)
5:  $\mathbf{y} \leftarrow \mathbf{X}\boldsymbol{\alpha}_r$ 
6:  $N \leftarrow$  number of rows of  $\mathbf{X}$ 
7:  $T \leftarrow$  number of columns of  $\mathbf{X}$ 
8:  $\mathbf{c} \leftarrow \left\lfloor \frac{1}{\sqrt{1+\kappa_2}} \mathbf{X}\mathbf{y} \right\rfloor$ 
9:  $p \leftarrow \max \mathbf{c}$ 
10:  $\mathcal{I} \leftarrow (1, \dots, N)$ 
11:  $\mathcal{A} \leftarrow \emptyset$ 
12:  $d \leftarrow \text{false}$ 
13:  $\mathbf{s} \leftarrow (\cdot)$ 
14:  $\hat{\boldsymbol{\lambda}}_r \leftarrow \mathbf{0}_N$ 
15:  $\hat{\mathbf{q}} \leftarrow \mathbf{0}_N$ 
16: while  $2p\sqrt{1+\kappa_2} > \kappa_{r,1} \wedge |\mathcal{A}| < M$  do
17:    $\hat{c} \leftarrow \max |\mathbf{x}_n \mathbf{y}|$  for  $n \in \mathcal{I}$  and  $\mathbf{x}_n \in \mathbf{X}$ 
18:    $\hat{n} \leftarrow \operatorname{argmax} |\mathbf{x}_n \mathbf{y}|$  for  $n \in \mathcal{I}$  and  $\mathbf{x}_n \in \mathbf{X}$ 
19:   if  $\neg d \wedge |\mathcal{A}| < N$  then
20:      $\mathcal{A} \leftarrow \mathcal{A} \cup \{\hat{n}\}$ 
21:      $\mathcal{I} \leftarrow \mathcal{I} \setminus \{\hat{n}\}$ 
22:      $\mathbf{s} \leftarrow \mathbf{s} \oplus \operatorname{sign}(\hat{c})$ 
23:     if  $|\mathcal{A}| = 1$  then
24:        $\mathbf{L} \leftarrow \sqrt{\mathbf{x}_n \mathbf{x}_n + \kappa_2} / (1 + \kappa_2)$ 
25:     else
26:       Update the lower Cholesky matrix  $\mathbf{L}$  by  $\mathbf{x}_{\hat{n}}$ 
27:     end if
28:   end if
29:   if  $d \vee |\mathcal{A}| = N$  then
30:      $d \leftarrow \text{false}$ 
31:   end if
32:    $\mathbf{g} \leftarrow \mathbf{s}$ 
33:    $\mathbf{g} \leftarrow \mathbf{v}_1$ , where  $\mathbf{v}_1$  is the solution to  $\mathbf{L}\mathbf{v}_1 = \mathbf{g}$ 
34:    $\mathbf{g} \leftarrow \mathbf{v}_2$ , where  $\mathbf{v}_2$  is the solution to  $\mathbf{L}'\mathbf{v}_2 = \mathbf{g}$ 
35:    $a \leftarrow 1/\sqrt{\mathbf{g}'\mathbf{s}}$ 
36:    $\mathbf{w} \leftarrow a\mathbf{g}$ 
37:    $\mathbf{u} \leftarrow \left( \frac{1}{\sqrt{1+\kappa_2}} \mathbf{X}[\mathcal{A}, \cdot]' \mathbf{w} \right) \oplus \left( \frac{\sqrt{\kappa_2}}{\sqrt{1+\kappa_2}} \mathbf{w} \right)$ 
38:    $\boldsymbol{\gamma} \leftarrow -1 \cdot (\hat{\boldsymbol{\lambda}}_r[\mathcal{A}] \oslash \mathbf{w})$ 
39:   if  $0 < \max \boldsymbol{\gamma}$  then
40:      $\tilde{\gamma} \leftarrow \max \boldsymbol{\gamma}$ 
41:      $n \leftarrow \operatorname{argmin} \gamma_i$  for  $\gamma_i \in \boldsymbol{\gamma}$ 
42:   else
43:      $\tilde{\gamma} \leftarrow \infty$ 

```

```

44:   end if
45:    $\hat{\gamma} \leftarrow \hat{c}/a$ 
46:   if  $|\mathcal{A}| < N$  then
47:      $\alpha \leftarrow \mathbf{X}[\mathcal{A}, \cdot] \mathbf{u}[1 : T] + \sqrt{\kappa_2} \mathbf{u}[(T+1) : |\mathcal{A}|]$ 
48:     for  $m \in (1, \dots, |\mathcal{I}|)$  do
49:        $h_1 \leftarrow (\hat{c} - c_{i_m})/(a - \alpha_m)$ , where  $i_m \in \mathcal{I}$ ,  $c_{i_m} \in \mathbf{c}$ , and  $\alpha_m \in \alpha$ 
50:        $h_2 \leftarrow (\hat{c} + c_{i_m})/(a + \alpha_m)$ , where  $i_m \in \mathcal{I}$ ,  $c_{i_m} \in \mathbf{c}$ , and  $\alpha_m \in \alpha$ 
51:       if  $0 < h_1 < \hat{\gamma}$  then
52:          $\hat{\gamma} \leftarrow h_1$ 
53:       end if
54:       if  $0 < h_2 < \hat{\gamma}$  then
55:          $\hat{\gamma} \leftarrow h_2$ 
56:       end if
57:     end for
58:   end if
59:    $\gamma^\dagger \leftarrow \min\{\tilde{\gamma}, \hat{\gamma}\}$ 
60:    $\boldsymbol{\lambda}^\dagger \leftarrow \hat{\boldsymbol{\lambda}}_r$ 
61:    $\hat{\boldsymbol{\lambda}}_r \leftarrow \hat{\boldsymbol{\lambda}}_r + \gamma^\dagger \mathbf{w}$ 
62:    $\hat{\mathbf{q}} \leftarrow \hat{\mathbf{q}} - \gamma^\dagger \mathbf{u}$ 
63:    $p^\dagger \leftarrow p - |\gamma^\dagger a|$ 
64:   if  $2p^\dagger \sqrt{1 + \kappa_2} > \kappa_{r,1}$  then
65:      $q_1 \leftarrow 2p^\dagger \sqrt{1 + \kappa_2}$ 
66:      $q_2 \leftarrow 2p \sqrt{1 + \kappa_2}$ 
67:      $\hat{\boldsymbol{\lambda}}_r \leftarrow \left( \frac{q_2 - \kappa_{r,1}}{q_2 - q_1} \hat{\boldsymbol{\lambda}}_r + \frac{\kappa_{r,1} - q_1}{q_2 - q_1} \boldsymbol{\lambda}^\dagger \right) (\sqrt{1 + \kappa_2})^{-1}$ 
68:      $p \leftarrow p^\dagger$ 
69:   else
70:     if  $\tilde{\gamma} < \hat{\gamma}$  then
71:        $d \leftarrow \text{true}$ 
72:        $\hat{\boldsymbol{\lambda}}_r[n] \leftarrow 0$ 
73:        $\mathcal{I} \leftarrow \mathcal{I} \cup \{n\}$ 
74:        $\mathbf{s} \leftarrow \mathbf{s} \ominus s_{|\mathcal{A}|}$ 
75:        $\mathcal{A} \leftarrow \mathcal{A} \setminus \{n\}$ 
76:       Downdate the lower Cholesky matrix  $\mathbf{L}$  by  $\mathbf{x}_n$ 
77:     end if
78:   end if
79: end while
80: return  $\hat{\boldsymbol{\lambda}}_r$ 

```

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